From Three Languages to Epistemic Loops: Representation, Sensory Coupling, and Autonomous Scientific Discovery

Manuscript type: Conceptual perspective / hypothesis and theory
Status: Working manuscript, not peer reviewed
Date: November 2026
Authorship and affiliations: To be determined before submission

Abstract

A recent account of machine intelligence proposes that human knowledge is created and communicated through three broad languages: natural language, mathematical formalism, and the arts. It further suggests that machines mastering these languages and generating knowledge autonomously could surpass human intellectual capacities. We examine this proposal as a productive organizing hypothesis rather than an exhaustive taxonomy or a sufficiency theorem. Its central omission is not simply an additional sensory channel. Rather, representation, perception, bodily regulation, action, and experimental intervention participate in coupled processes through which knowledge is constructed and tested. We synthesize insights from symbol grounding, embodied and enactive cognition, predictive processing, multisensory integration, neurocognitive entrainment, and automated scientific discovery into a representational-sensorimotor-epistemic loop. In this framework, natural language, mathematics, and the arts supply partially overlapping forms of representation and communication; sensory coupling constrains and updates models; interventions generate discriminating evidence; and explicit validation converts candidate interpretations into warranted knowledge. Sensory stimulation is therefore both an object of scientific explanation and a controllable experimental intervention, not a peripheral application. A formal state-space formulation and five falsifiable propositions specify how multimodal and interactive machine systems might be compared with purely representational systems, and how adaptive sensory experiments could be evaluated without conflating stimulus-locked neural responses with cognitive benefit. We argue that the emerging machine-generated-knowledge paradigm should be assessed by independently verified causal discoveries, calibration, reproducibility, and transfer, not merely processing speed or linguistic fluency. The account neither requires machine embodiment for every scientific task nor presupposes that sensory stimulation increases general intelligence.

Keywords: machine-generated knowledge; three languages; symbol grounding; embodied cognition; enactivism; sensory stimulation; multisensory integration; active learning; autonomous science; epistemology

1. Introduction: the epistemic promise and limits of the three-language thesis

Oliveira Jr. and Oliveira [1] argue that human knowledge is fundamentally created and transmitted through natural languages, mathematical formalisms, and the arts. Their larger claim concerns a possible transition from human-generated to machine-generated knowledge, described in relation to a proposed fifth paradigm of science [1,2]. The historical sequence on which this framing builds is familiar: empirical observation, mathematized theory, computational simulation, and data-intensive scientific discovery [3]. Increasing automation may change not only scientific instruments but also which agents formulate hypotheses and decide what evidence should be sought.

The three-language thesis identifies an important feature of intellectual work: propositions, formal relationships, and aesthetically or perceptually organized expressions are portable across individuals and generations. Yet the thesis makes two distinct claims that must be separated. A *taxonomic claim* holds that these three domains usefully classify major forms of human knowledge representation. A *sufficiency claim* holds that mastery of them is enough for an entity to perform every human activity. The first can function as a heuristic. The second does not follow without additional assumptions about perception, action, goals, causal competence, material resources, social coordination, and the assessment of truth.

Our intervention is not to add "sensory language" as a fourth item. Sensory phenomena already traverse the proposed categories: speech is heard or read; mathematical concepts are often learned through diagrams, gesture, and manipulation; the arts can engage hearing, vision, touch, proprioception, and movement. Conversely, not every sensory contingency is naturally a language. The more consequential move is to distinguish *systems of representation* from the *couplings and procedures by which representational systems gain, revise, and test their reference to the world*.

We develop an integrative framework with four coupled functions: (i) representation and communication, (ii) sensory and interoceptive observation, (iii) action and intervention, and (iv) epistemic evaluation and memory. These are analytical functions, not a claim that the mind or an artificial agent contains four discrete modules. We then examine sensory stimulation as a particularly revealing case because a designed acoustic, visual, tactile, or multimodal pattern can simultaneously carry meaning, evoke perception, perturb physiological dynamics, and create an opportunity for causal inference.

This is a conceptual article. It reports no new behavioral experiments, clinical outcomes, trained models, or quantitative meta-analysis. Its contribution lies in a clarified theory, explicit alternative explanations, a minimal formalization, and empirical predictions that can fail.

2. What counts as a language, and what counts as knowing?

2.1. Three representational families, not three exhaustive channels

Natural language supports open-ended discourse, instruction, narration, negotiation, and reporting of observations. Mathematics emphasizes compositional, formally constrained relations, allowing predictions, proofs, and consistency checks. Artistic practices use organized perceptual, temporal, spatial, and symbolic structures whose functions can include expression, interpretation, social coordination, and exploration. These families overlap: a musical score is mathematical and conventional; architecture relies on material affordances, geometry, and aesthetic judgments; a scientific visualization can be representational and artistic at once.

Consequently, "language" must be used broadly, as a family-resemblance term rather than as a formal equivalence between linguistic syntax, mathematical notation, and artistic experience. There is no demonstrated theorem that these three families form a complete or minimal basis for all knowledge. Nor is it established that their mastery entails the ability to carry out any human act. A skilled theorem prover can establish claims without sensory intervention; a robot can learn useful sensorimotor policies without explicit propositions. These cases complicate both purely linguistic and exclusively embodied accounts.

The apparent plausibility of representational sufficiency often comes from collapsing three different achievements: producing well-formed expressions, connecting those expressions to a domain, and establishing that the relevant claims are true or effective. The distinction is central to the symbol-grounding problem [4] and to contemporary debates over the relation between linguistic form and meaning [8]. Text can encode information gathered from the world by other agents. Thus, a system trained only on text can acquire extensive *indirect* worldly knowledge. The open question is not whether it can acquire any knowledge, but how it can detect when inherited descriptions are incomplete, misleading, or no longer applicable.

2.2. Perception is neither a passive pipe nor an optional decorative layer

Cognitive systems do not merely receive uninterpreted facts and translate them into words. Perception selects, combines, and interprets signals, often using prior expectations and bodily contingencies. Accounts of grounded cognition [5], enaction [6], behavior-based robotics [7], and predictive processing [10,11] differ substantially, but collectively challenge a sharp boundary between abstract cognition and sensorimotor engagement.

These theories should not be conflated. Grounded cognition emphasizes the contribution of modal simulation and situated action to conceptual processing [5]. Enactive approaches emphasize organism-environment relations and the role of lived activity [6]. Brooks's robotics program questions the need for centralized symbolic representations in some intelligent behavior [7]. Predictive-processing models emphasize generative predictions and prediction errors, but do not establish by themselves that any particular philosophical account of embodiment

is correct [10,11]. Their common relevance here is a more modest proposition: perceptual and action systems can materially constrain what an agent learns and how it generalizes.

Multisensory perception offers a concrete example. Vision, audition, and touch can be combined in ways that depend on reliability, timing, prior information, and task demands [12]. Multisensory learning can outperform some single-modality training procedures, but not universally or independently of experimental design [13]. The senses are not simply additional files to be ingested. They provide correlated, partially independent measurements of underlying causes, and sometimes reveal that an existing model is wrong.

Interoception also matters. Bodily signals relevant to autonomic regulation, affect, fatigue, and arousal influence cognitive context; theoretical accounts of active interoceptive inference connect such regulation with prediction and action [26]. For artificial science, the analogous requirement is not a biological interior, but access to operational state: instrument calibration, uncertainty, resource constraints, sensor integrity, and the capacity to act on observed failures. This is a functional analogy, not evidence that machines possess biological affect or phenomenal experience.

3. A representational-sensorimotor-epistemic loop

We propose the *representational-sensorimotor-epistemic loop* (RSEL) as a descriptive framework, not a newly discovered biological law. Its central unit is a cycle of model construction and correction rather than a fixed inventory of language types.

Represent. An agent encodes hypotheses, explanations, constraints, procedures, and observations in natural language, mathematics, visual structures, sound, code, diagrams, or other media. The original three families are useful here, without excluding other representational practices.

Observe. An agent receives sensory or instrument-mediated evidence, including records generated by other agents. Observation is structured by sensors, sampling, calibration, context, and prior models. A microscope, questionnaire, telescope, simulation, or human participant may serve as a source of observations. Direct bodily embodiment is one possible route, not a necessity for every scientific discovery.

Intervene. An agent chooses actions that can modify a task or reveal its structure. Kirsh and Maglio [9] distinguished pragmatic actions, which advance an external goal, from epistemic actions, which improve knowledge. Experiments, instrument adjustments, targeted queries, robotic manipulation, and structured sensory stimulation can all be epistemic actions when selected to discriminate hypotheses.

Evaluate and retain. The resulting claims must survive comparison with controls, uncertainty analysis, independent measurements, and replication. A paper,

model prediction, or plausible narrative is not automatically validated knowledge. Reliable memory requires source provenance and mechanisms to correct, supersede, or discard false claims.

These functions are linked in both directions. A hypothesis changes what is measured. Evidence changes which representations are retained. An intervention changes the available evidence, and its observed effect changes future intervention choices. Artistic structures can be the subject of perception, the vehicle for a representation, and a manipulable experimental input in the same cycle. This is precisely why sensory stimulation should be integrated into the epistemic picture from the start.

A diagrammatic summary is:

Representation and model -> predicted observations -> sensory or instrument-mediated evidence -> intervention and comparison -> validated or revised model -> representation and model.

The arrow ordering is schematic: real systems often monitor, intervene, and update concurrently. The epistemic advance lies not in the loop as a metaphor, which has many antecedents, but in specifying its components and evaluating where machine autonomy can and cannot replace human decisions.

3.1. Minimal mathematical description

Let z_t be the relevant latent state of an organism, instrument, or environment, θ denote unknown dynamical parameters, u_t an intervention, c_t context, and y_t observable data. A generic state-space model is

$$z_{t+1} = f_\theta(z_t, u_t, c_t) + \epsilon_t, \qquad y_t = g_\theta(z_t, c_t) + \nu_t.$$

An agent maintains a belief $p(\theta, z_t \mid D_t)$ from a history D_t of observations, interventions, and metadata. Its next intervention may be chosen according to

$$u_t^* \in \arg\max_{u \in \mathcal{U}_{\text{safe}}} \left[\mathbb{E}\{ I(\theta; Y_{t+1} \mid D_t, \text{do}(u)) \} - \lambda C(u) \right],$$

where I is conditional mutual information, $\text{do}(u)$ denotes a controlled intervention, C aggregates experimental cost and burden, λ determines the cost tradeoff, and $\mathcal{U}_{\text{safe}}$ constrains acceptable interventions. The expression is an *idealized objective*, not a claim that real AI systems correctly infer causal effects by applying it. The expectation is only as good as the assumed model, measurement process, and intervention design. If the causal structure is misspecified, information optimization can efficiently produce misleading certainty. Classical information-based experimental design and Bayesian optimization offer related technical precedents [22,23].

In computational science, u might select a simulation parameter; in robotic sci-

ence, a reagent or manipulation; in neuroscience, a constrained stimulus protocol. Sensory stimulation is not privileged mathematically. It is scientifically distinctive because the intervention may change both the observations accessible to the participant and the participant's physiological or cognitive state. As a result, stimulus effects on reporting, attention, motivation, learning, or arousal must be separated rather than inferred from an aggregate behavioral score.

4. Sensory stimulation as representation, evidence, and intervention

4.1. The same stimulus can have different epistemic roles

Consider a rhythmic auditory sequence. Its temporal arrangement can encode information, as in musical meter or a signaling protocol. It can be perceived and predicted by a listener, providing evidence about temporal cognition. It can also be manipulated experimentally to test whether specific changes in rhythmic structure alter neural activity or task performance.

Thus, sound is not simply the artistic "third language" in another form. Acoustic structure is simultaneously a physical signal, an organized perceptual object, and a potential intervention into an organism-environment system. Similar arguments hold for visual temporal patterns, haptic stimulation, guided movement, breathing cues, and multimodal synchronization. The role of each must be operationally defined for the particular experiment.

Music illustrates the integration particularly clearly. Its mathematical features include temporal intervals, spectral structure, amplitude envelopes, and phase relations; its artistic dimensions include interpretation, expectation, culturally mediated meaning, and affect. Behavioral and physiological responses depend on learning, preference, task, and context. One cannot infer a unique mental state or scientific truth from a single acoustic frequency. What is promising is a structured and repeatable space of stimuli whose causal effects can be tested, not a dictionary translating frequencies directly into cognitive abilities.

4.2. What neuroscience establishes, and what it does not

Research on auditory and visual temporal regularities shows that neural responses can reflect stimulus timing, attention, and expectations. Lakatos and colleagues [15] reported rhythmic modulation of sensory cortical activity associated with attentional selection. Nozaradan and colleagues [14] investigated neural signatures of musical beat and meter. Reviews of rhythmic stimulation address entrainment mechanisms and their interpretive limitations [16,17].

Critically, a neural response at the stimulus frequency is *not by itself* proof of entrainment of an autonomous endogenous oscillator. Repeated evoked responses can produce similar spectral patterns, a distinction emphasized by Zoefel and colleagues [16]. Nor does frequency-specific neural activity establish improved reasoning, memory, or general intelligence.

One instructive human experiment used closed-loop auditory stimulation timed to sleep slow oscillations and reported effects on sleep-related electrophysiology and declarative-memory consolidation in the studied conditions [18]. It supports the feasibility of targeted, state-contingent sensory intervention. It does not support universal cognitive enhancement, arbitrary frequency-to-function mappings, or broad claims about all stimulation technologies. Some animal gamma-entrainment studies have generated mechanistic hypotheses [27], but species, protocol, and disease-model differences preclude direct generalization to human clinical efficacy.

A useful hierarchy is:

1. **Physical fidelity:** Did the device deliver the intended signal?
2. **Perceptual engagement:** Did the participant detect, interpret, or otherwise process it?
3. **Neurophysiological response:** Is there a reliable, artifact-controlled physiological difference?
4. **Functional effect:** Is there a preregistered, reproducible behavioral change relative to a suitable control?
5. **Generalization:** Does the benefit persist, transfer, and hold across populations without unacceptable adverse effects?

Evidence at one level does not automatically establish the next. The higher levels require stronger experiments, and each can fail even when lower levels are established.

4.3. Why multisensory stimulation matters to machine-produced knowledge

A scientific agent studying perception cannot assume a one-to-one mapping between physical input and subjective or behavioral response. Multisensory integration depends on timing, reliability, and learned regularities [12,13]. Interoceptive state and task engagement may further modify the response [26]. Hence a multimodal intervention should be understood as a *context-sensitive test* of a model rather than a universally optimal stimulation recipe.

The same logic applies in reverse: models trained from multiple independent measurement channels can sometimes identify confounds and detect model mismatch that cannot be distinguished using language descriptions alone. Yet adding modalities also adds synchronization errors, missing-data mechanisms, participant burden, and new pathways to leakage. More signals are valuable only when they increase identifiable information about the intended causal question.

5. What the fifth paradigm would have to demonstrate

The concept of machine-generated knowledge precedes current generative AI and has been articulated explicitly by Oliveira Jr. and Oliveira [2]. Earlier robotic systems demonstrated restricted cycles of hypothesis generation and experimental

testing. The robot scientist Adam, for example, generated functional-genomics hypotheses and tested them through automated experiments [19]. More recently, the AI Scientist pipeline automated several stages of machine-learning research, including ideation, coding, experimentation, and manuscript drafting [20]. Its authors also documented errors, fragile rigor, and a limited peer-review success rate. Independent methodological critiques of autonomous AI science identify additional risks of flawed evaluation and hidden research choices [21].

These results support a careful conclusion: increasingly autonomous scientific workflows exist, including examples that operate beyond mere text generation. They do *not* yet establish generalized, autonomous, reliably self-correcting science at arbitrary domain breadth. In particular, a scientifically competent system must be judged by whether its claims survive external testing, not by how convincingly it states those claims.

The proposed fifth paradigm is best characterized as a shift in the allocation of epistemic labor. Machines may select questions, build instruments, acquire observations, select interventions, fit models, and generate explanations; humans and institutions may still specify objectives, oversee risky experiments, and adjudicate consequential claims. Autonomy is therefore multidimensional, not a binary state. A machine scientist can be highly autonomous in in-silico optimization and dependent on human-operated instruments for physical measurements.

This perspective also resolves an apparent tension between language-focused AI and embodied theories. A machine can contribute valid knowledge without possessing a human body. Mathematical proof, archival inference, and software verification may require no novel physical act. For empirical claims, however, *someone or something* in the evidential chain must connect hypotheses with observations and discriminating interventions. Grounding may be direct, delegated, distributed, or historical. The scientifically relevant question is whether the chain is reliable and auditable, rather than whether all cognition must be anatomically embodied. The related cognitive-science account of Lake and colleagues stresses causal world models, intuitive physics, and compositionality rather than pattern recognition alone [25].

6. Testable propositions and discriminating experiments

The value of an integrative framework depends on whether it yields predictions beyond general statements that "the senses matter." We propose five tests. The propositions concern specific tasks and populations, not a universal law of cognitive enhancement.

P1: Closed-loop access improves learning under contingent dynamics. For an initially unknown physical or simulated process whose parameters change over time, agents permitted calibrated observations and chosen interventions should outperform otherwise matched agents restricted to static textual descriptions, *when* the task requires new information unavailable in the descriptions. Outcome: externally scored predictive calibration and intervention success on held-out con-

ditions. A null result would constrain the claim that online sensorimotor feedback is advantageous in that task. This comparison must control model capacity, compute, observation budgets, and access to historical data.

P2: Epistemic interventions outperform passive observations under identifiable uncertainty. For a task with distinct causal models that make identical predictions on passive training data but different predictions under interventions, an agent able to select appropriate tests should resolve the ambiguity more efficiently. Outcome: accuracy of causal identification and verified out-of-distribution predictions per unit experimental cost. If observational data already identify the model, no advantage is predicted. The causal setup must be transparent, with genuinely blinded evaluation.

P3: Sensory effects cannot be inferred from stimulus-frequency locking alone. In randomized rhythmic-stimulation experiments, reliable stimulus-frequency responses need not coincide with reliable preregistered behavioral benefits. This is a dissociation prediction, not a prediction that all benefits are absent. Outcome: joint distributions of stimulus-locked responses, evoked-response controls, behavioral effect sizes, and individual heterogeneity. A robust one-to-one relation across independently replicated protocols would challenge the proposed cautious separation.

P4: Adaptive interventions offer conditional rather than universal gains. A closed-loop sensory protocol that uses a prespecified physiological marker may outperform a matched open-loop protocol when the marker is reliable and predictive of responsiveness. The advantage should weaken when feedback is noisy or temporally uninformative. Outcomes: registered primary task or physiological measures, adverse events, and algorithm-to-algorithm comparisons. Personalization must be evaluated on unseen participants or temporally held-out sessions to avoid optimistic adaptation bias.

P5: Independent validation changes the apparent ranking of machine scientists. Rankings based on manuscript fluency, internal reviewers, or self-reported discoveries should diverge from rankings based on prospective reproduction and independently generated tests when failure modes are present. Outcomes: prospective replication rate, error calibration, correct negative findings, research cost, and the proportion of claims surviving independent scrutiny. If rankings remain stable across preregistered independent checks, the proposed gap is smaller than expected.

These propositions can be investigated separately. None depends on demonstrating artificial consciousness, general intelligence enhancement, or a technological singularity.

6.1. An illustrative experimental design

As a concrete example, consider a laboratory study of auditory temporal patterns and sustained attention. The study would preregister a primary behavioral endpoint, inclusion criteria, exclusion criteria, sample-size rationale, analysis plan,

acceptable sound exposure, and stopping conditions. Participants could complete randomized, counterbalanced sessions with (a) an experimentally defined rhythmic pattern, (b) a spectrally and energetically matched non-target temporal control, and (c) a second active control to distinguish rhythm-specific effects from novelty or preference. If EEG is available, artifact-controlled neural outcomes should be designated as secondary or co-primary *before* data collection.

A fixed protocol and a feedback-driven protocol could be compared at matched cumulative exposure and participant burden. The adaptive algorithm's allowable changes, training data, optimization objective, and safety constraints must be frozen before confirmatory testing. Outcome assessors should be blinded where feasible; subjective expectancy and musical preferences should be measured. Analysis should report effect sizes with uncertainty, corrections for multiplicity, individual differences, and null results. Replication on separate participants or in another laboratory would be essential before generalizing.

This is a proposed study, not an experiment performed for this manuscript. No benefit is assumed in advance. A well-powered null result would be informative about the corresponding intervention and measurement configuration.

7. Quantitative and conceptual cautions

The force of the original proposal does not depend on a simple speed multiplier between biological and computational systems. Such comparisons need a common operational task and matched definitions of processing. Token throughput is not comprehension; corpus size is not validated understanding; accelerated literature synthesis is not automatically accelerated discovery. Computational scale may substantially expand search and synthesis while leaving experimental bottlenecks and epistemic error intact.

Two illustrative arithmetic claims in the source essay [1] should not be reproduced uncritically. Reading 7,000 to 10,000 words per day for 30 years corresponds to approximately 76.7 to 109.6 million words, rather than one billion. Likewise, 10^{13} seconds corresponds to about 317,000 years, not one million. These corrections are minor relative to the theory but illustrate why scientific arguments about speed require dimensional checks.

There are further limits to the present synthesis:

- **Taxonomic ambiguity.** The proposed languages are heterogeneous, overlapping cultural practices, not independent axes established by factor analysis or information theory.
- **Theoretical pluralism.** Embodied cognition, enactivism, ecological approaches, and predictive processing are not interchangeable and make different commitments. Our loop is a compatible analytical scaffold, not empirical confirmation of one theory.
- **No universal embodiment requirement.** Disembodied mathematics, simulation, archival science, and tool-mediated observation are valid counterexamples to claims that firsthand sensing is necessary for all knowledge.

- **Causality and identifiability.** A closed-loop algorithm can optimize a proxy while leaving the causal target unidentified. Randomization alone does not repair every measurement or modeling problem.
- **Neuroscientific specificity.** Physiological phase locking may reflect evoked responses, attention, or artifacts. Behavioral effects can arise from expectations, arousal, fatigue, or acoustic preferences rather than the hypothesized mechanism.
- **Cross-domain transfer.** Changes in a narrowly defined task do not establish general cognitive enhancement or durable neuroplastic effects.
- **Social epistemology.** Instruments, institutions, reviewers, funding, value judgments, and communities shape what counts as evidence. Scientific credibility is not a direct consequence of individual agent competence.
- **Consciousness.** The framework is functional and epistemic. It neither answers whether an artificial system is conscious nor assumes that manipulating sensory inputs is sufficient to modify subjective experience in a specific way.
- **Potential dual use and oversight.** Adaptive experiments involving people require competent ethics review, informed consent, privacy safeguards, calibrated exposure limits, and the ability to halt an intervention.

Theoretical adequacy also requires an account of failure. We should expect the framework to add little value when evidence is already complete, causal variables are inaccessible, measurement is poor, or new interventions are too costly or unsafe to run.

8. Implications for research infrastructure

If machine systems are to reason about experimental sensory protocols, records must differentiate *what was intended*, *what was actually delivered*, and *what was observed*. A useful interoperable minimum includes:

Layer	Example content	Epistemic function
Hypothesis and representation	claim, predicted effect, competing causal models, formal constraints	Specifies what could be learned
Protocol	stimulus modality, waveform, timing, intensity, assignment, controls, intended endpoints	Specifies an intended intervention
Execution	device configuration, calibration, timestamps, deviations, participant context	Records the intervention as delivered

Layer	Example content	Epistemic function
Observation and event	physiological traces, behavioral outcomes, reports, adverse events	Provides data, not yet conclusions
Evaluation	analysis version, model fit, uncertainty, preregistration, replication status	Converts candidate findings into auditable evidence

The distinction is similar in spirit to the FAIR principles for scientific data stewardship [24]. It is not tied to any particular commercial application or standard. Interoperability would help autonomous or human researchers compare protocols without mistaking labels for identical physical stimuli or treating a recorded execution as a verified causal finding. Persistent identifiers, ontologies, machine-readable metadata, and open analysis pipelines may make machine-assisted research easier to audit. They do not themselves guarantee semantic validity or reproducibility.

This architecture is particularly valuable for multimodal experimental science because nominally identical stimuli may differ at the participant due to equipment, transmission, timing, calibration, or environment. A machine scientist operating on poorly described stimulus labels can only give the appearance of reproducibility.

9. Discussion: intelligence, knowledge, and the distribution of epistemic agency

The three-language account [1] is useful as an invitation to think beyond language models understood narrowly as text predictors. Its limitation is an implicit leap from command of expressive forms to universal competence. The present synthesis replaces that leap with a conditional hypothesis: machine scientific capabilities should expand as systems become better at integrating symbolic and non-symbolic representations with trustworthy observation, discriminating intervention, and independent validation.

There are two important counterarguments. First, increasingly capable representation learning may internalize enough structure from human-produced descriptions, videos, simulations, and scientific archives to operate successfully without direct sensory contact. This is a serious possibility and a reminder that grounding can be inherited or mediated. Second, embodied action can be expensive and noisy, whereas a well-specified mathematical model may generate more reliable answers than exploratory interaction. Neither objection defeats the loop account. Both show that the relative value of sensing and acting must be established task by task rather than asserted in advance.

The proposed framework also changes the interpretation of human-machine com-

parison. Human intelligence is not homogeneous: perception, language, sensorimotor expertise, scientific reasoning, memory, social cooperation, and institutional validation are partly separable capacities. Machines may exceed human performance on some dimensions long before they match others. Conversely, superior throughput or broad linguistic proficiency cannot alone establish epistemic superiority. The relevant comparison is between independently verified accomplishments under stated resources and uncertainty.

Sensory stimulation supplies a useful test case because it makes the reciprocal character of knowledge especially visible. An agent designs a stimulus using mathematical and artistic structure, delivers it to an organism, measures context-dependent responses, and revises its model. Knowledge about the organism guides the next intervention, while the intervention changes the organism's observable state. At every step, representation and causal engagement coexist. This is not a special addendum to language; it is an instance of how scientific inquiry operates.

One conceivable longer-term development is a feedback process in which AI advances experimental design, scientific discoveries improve machine capabilities, and some resulting interventions help humans learn or perform particular tasks. This possibility is worth testing, but neither a general intelligence increase from sensory stimulation nor a self-sustaining recursive explosion of AI capability follows from the evidence reviewed here.

10. Conclusion

Natural language, mathematics, and the arts provide a compelling, though non-exhaustive, sketch of how knowledge is expressed and shared. Scientific intelligence also depends on how representations are connected to observation, action, and independent tests. Sensory processing and designed sensory stimulation are intrinsic examples of these connections: the former constrains models of the world; the latter can be both meaningful expression and a controlled experimental input.

The central proposal is therefore not a fourth language, but an epistemic loop. In this view, a machine scientist is distinguished less by fluent expression or computational speed than by its capacity to acquire warranted, revisable knowledge through reliable observation, judicious intervention, transparent uncertainty, and reproducible validation. The fifth paradigm will be scientifically compelling to the extent that its discoveries meet these criteria.

Declarations

Article type: Conceptual perspective with testable propositions. No new experimental data, participants, or human-subject interventions are reported.
Empirical status: Literature-based synthesis; no systematic review or meta-analysis was performed.

Authorship: Authors, affiliations, and contribution statements require confirmation before academic submission. The draft was prepared with AI assistance and must undergo author-led source checking and substantive scientific review.
Funding and conflicts of interest: Not declared at draft stage; to be completed accurately by the submitting author(s).
Data availability: No new data were generated or analyzed. All cited sources are linked below.
Code availability: No new research software is required for the theoretical claims.
Research ethics: Proposed human sensory-stimulation studies would require appropriate prospective oversight and consent.

References

1. Oliveira Jr., O. N., & Oliveira, M. C. F. (2026). *Por que máquinas inteligentes poderão superar os seres humanos?* ComCiência, 3 June. https://www.comciencia.br/porque-maquinas-inteligentes-poderao-superar-os-seres-humanos/
2. Oliveira Jr., O. N., & Oliveira, M. C. F. (2022). Materials discovery with machine learning and knowledge discovery. *Frontiers in Chemistry, 10*, 930369. https://doi.org/10.3389/fchem.2022.930369
3. Hey, T., Tansley, S., & Tolle, K. (Eds.). (2009). *The Fourth Paradigm: Data-Intensive Scientific Discovery.* Microsoft Research. https://www.microsoft.com/en-us/research/publication/the-fourth-paradigm-data-intensive-scientific-discovery/
4. Harnad, S. (1990). The symbol grounding problem. *Physica D, 42*(1-3), 335-346. https://doi.org/10.1016/0167-2789(90)90087-6
5. Barsalou, L. W. (2008). Grounded cognition. *Annual Review of Psychology, 59*, 617-645. https://doi.org/10.1146/annurev.psych.59.103006.093639
6. Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience.* MIT Press. https://mitpress.mit.edu/9780262220422/embodied-mind/
7. Brooks, R. A. (1991). Intelligence without representation. *Artificial Intelligence, 47*(1-3), 139-159. https://doi.org/10.1016/0004-3702(91)90053-M
8. Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. *Proceedings of ACL 2020*, 5185-5198. https://doi.org/10.18653/v1/2020.acl-main.463
9. Kirsh, D., & Maglio, P. (1994). On distinguishing epistemic from pragmatic action. *Cognitive Science, 18*(4), 513-549. https://doi.org/10.1207/s15516709cog1804_1
10. Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences, 36*(3), 181-204. https://doi.org/10.1017/S0140525X12000477
11. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience, 11*, 127-138. https://doi.org/10.1038/nrn2787
12. Ernst, M. O., & Bülthoff, H. H. (2004). Merging the senses into a robust percept. *Trends in Cognitive Sciences, 8*(4), 162-169. https://doi.org/10.1016/j.tics.2004.02.0

13. Shams, L., & Seitz, A. R. (2008). Benefits of multisensory learning. *Trends in Cognitive Sciences, 12*(11), 411-417. https://doi.org/10.1016/j.tics.2008.07.006
14. Nozaradan, S., Peretz, I., Missal, M., & Mouraux, A. (2011). Tagging the neuronal entrainment to beat and meter. *Journal of Neuroscience, 31*(28), 10234-10240. https://doi.org/10.1523/JNEUROSCI.0411-11.2011
15. Lakatos, P., Karmos, G., Mehta, A. D., Ulbert, I., & Schroeder, C. E. (2008). Entrainment of neuronal oscillations as a mechanism of attentional selection. *Science, 320*(5872), 110-113. https://doi.org/10.1126/science.1154735
16. Zoefel, B., ten Oever, S., & Sack, A. T. (2018). The involvement of endogenous neural oscillations in the processing of rhythmic input: more than a regular repetition of evoked neural responses. *Frontiers in Neuroscience, 12*, 95. https://doi.org/10.3389/fnins.2018.00095
17. Thut, G., Schyns, P. G., & Gross, J. (2011). Entrainment of perceptually relevant brain oscillations by non-invasive rhythmic stimulation of the human brain. *Frontiers in Psychology, 2*, 170. https://doi.org/10.3389/fpsyg.2011.00170
18. Ngo, H.-V. V., Martinetz, T., Born, J., & Mölle, M. (2013). Auditory closed-loop stimulation of the sleep slow oscillation enhances memory. *Neuron, 78*(3), 545-553. https://doi.org/10.1016/j.neuron.2013.03.006
19. King, R. D., et al. (2009). The automation of science. *Science, 324*(5923), 85-89. https://doi.org/10.1126/science.1165620
20. Lu, C., et al. (2026). Towards end-to-end automation of AI research. *Nature, 651*, 914-919. https://doi.org/10.1038/s41586-026-10265-5
21. Luo, Z., Kasirzadeh, A., & Shah, N. B. (2026). The more you automate, the less you see: Hidden pitfalls of autonomous AI scientists. *Proceedings of the National Academy of Sciences, 123*(41), e2610214123. https://doi.org/10.1073/pnas.2610214123
22. MacKay, D. J. C. (1992). Information-based objective functions for active data selection. *Neural Computation, 4*(4), 590-604. https://doi.org/10.1162/neco.1992.4.4.590
23. Shahriari, B., Swersky, K., Wang, Z., Adams, R. P., & de Freitas, N. (2016). Taking the human out of the loop: a review of Bayesian optimization. *Proceedings of the IEEE, 104*(1), 148-175. https://doi.org/10.1109/JPROC.2015.2494218
24. Wilkinson, M. D., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data, 3*, 160018. https://doi.org/10.1038/sdata.2016.18
25. Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences, 40*, e253. https://doi.org/10.1017/S0140525X16001837
26. Seth, A. K., & Friston, K. J. (2016). Active interoceptive inference and the emotional brain. *Philosophical Transactions of the Royal Society B, 371*(1708), 20160007. https://doi.org/10.1098/rstb.2016.0007
27. Iaccarino, H. F., et al. (2016). Gamma frequency entrainment attenuates amyloid load and modifies microglia. *Nature, 540*, 230-235. https://doi.org/10.1038/nature20587